

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I T. Karimulla / 192465046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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[Signature]

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Given Network : 192.168.100.0/24

(a) School of Computing - /25

- Network Address : 192.168.100.0
- Subnet Mask : 255.255.255.128
- First Host : 192.168.100.1
- Last Host : 192.168.100.126
- Broadcast Address : 192.168.100.127
- Usable Hosts : 126

(b) School of Electronics - /26

- Network Address : 192.168.100.128
- Subnet Mask : 255.255.255.192
- First Host : 192.168.100.129
- Last Host : 192.168.100.190
- Broadcast Address : 192.168.100.191
- Usable Hosts : 62

(c) School of Mechanical Engineering - /27

- Network Address : 192.168.100.192
- Subnet Mask : 255.255.255.224
- First Host : 192.168.100.193
- Last Host : 192.168.100.222
- Broadcast Address : 192.168.100.223
- Usable Hosts : 30

Remaining Unused Range

192.168.100.224 - 192.168.100.255 (32 IP addresses)

② Given Network: 10.10.0.0/16

Software Development - /24

- Network: 10.10.0.0
- First Host: 10.10.0.1
- Last Host: 10.10.0.254
- Broadcast: 10.10.0.255
- Usable Hosts: 254

Quality Assurance - /25

- Network: 10.10.1.0
- First Host: 10.10.1.1
- Last Host: 10.10.1.126
- Broadcast: 10.10.1.127
- Usable Hosts: 126

Human Resources - /26

- Network: 10.10.1.128
- First Host: 10.10.1.129
- Last Host: 10.10.1.190
- Broadcast: 10.10.1.191
- Usable Host: 62

First Host: 10.10.1.193

Last Host: 10.10.1.222

Broadcast: 10.10.1.223

Usable Hosts: 30

Remaining Unused Range

10.10.1.224 - 10.10.255.255

③ IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

1. Compressed Notation

2001:db8:abcd::/64

2. Expanded Notation

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

3. Network Prefix

2001:db8:abcd:0000::/64

4. Number of Host Addresses

$2^{64} = 18,446,744,073,709,551,616$ host addresses

5. Total Possible Host Addresses

18,446,744,073,709,551,616 (2^{64})

④ Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45

Answers:

1. Destination Subnet: 192.168.2.0/24
2. Matching Routing Entry: 192.168.2.0/24
3. Next-Hop interface: the router forwards the packet through the interface connected to the 192.168.2.0/24 network
4. Default Route: 0.0.0.0/0

⑤ Administration LAN

- Network Address: 192.168.10.0
- Subnet Mask: 255.255.255.192 (/26)
- First Host: 192.168.10.1
- Last Host: 192.168.10.62
- Broadcast Address: 192.168.10.63

Finance LAN

- Network Address: 192.168.10.64
- Subnet Mask: 255.255.255.192
- First Host: 192.168.10.65
- Last Host: 192.168.10.125
- Broadcast Address: 192.168.10.127

Suggested IP Addresses

Administration LAN

- PC1: 192.168.10.2
- PC2: 192.168.10.3
- PC3: 192.168.10.4

Finance LAN

- PC1: 192.168.10.66
- PC2: 192.168.10.67
- PC3: 192.168.10.68

Default Gateway:

- Administration LAN: 192.168.10.1
- Finance LAN: 192.168.10.65

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